PAGE 1 - Cell membrane

The cell membrane is a phospholipid bilayer that controls which substances enter and leave the cell. Embedded transport proteins move ions such as sodium and potassium across the membrane. Small nonpolar molecules like oxygen diffuse directly through the bilayer.

PAGE 2 - Mitochondria

Mitochondria produce most of the cell's ATP through cellular respiration. Glucose is first split during glycolysis in the cytoplasm, and the citric acid cycle then runs inside the mitochondrial matrix. Aerobic respiration yields about 38 ATP per glucose molecule in older textbook estimates.

PAGE 3 - Photosynthesis

Chloroplasts capture light energy using chlorophyll pigments. The light-dependent reactions in the thylakoid membranes split water and release oxygen. The Calvin cycle in the stroma fixes carbon dioxide into sugar using ATP and NADPH.

PAGE 4 - Cell division

Mitosis produces two genetically identical daughter cells and has four phases: prophase, metaphase, anaphase and telophase. During metaphase the chromosomes line up along the cell's equator. Meiosis instead produces four genetically different gametes with half the chromosome number.

PAGE 5 - Enzymes

Enzymes are proteins that speed up reactions by lowering the activation energy. Each enzyme has an active site that binds a specific substrate. High temperatures can denature an enzyme so that its active site no longer fits the substrate.

PAGE 6 - Revision checklist

Before the unit test on Friday, revise membrane transport, compare respiration with photosynthesis, draw the phases of mitosis, and explain why denatured enzymes stop working. The unit test lasts 45 minutes.